

Geometric Dynamic Location Theory: Temporal Experience and Inter-Configuration Correlations from Quantum State Space Curvature

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Abstract—We present a geometric formulation of Dynamic Location Theory (DLT) in which temporal experience and weak inter-configuration correlations emerge from the intrinsic curvature of quantum configuration space. Treating the projective Hilbert space of pure states as a Riemannian manifold endowed with the Fubini–Study metric, we derive a scalar coupling γ as a geometric invariant rather than a phenomenological parameter. The coupling is shown to be dimensionally consistent, Planck suppressed, and dependent on many-body configuration space curvature. Explicit calculations are provided for single qubits, composite systems, and realistic atomic clock models. The resulting predictions are consistent with current experimental bounds and yield falsifiable scaling laws with system size and entanglement depth.

Index Terms—timeless quantum mechanics, block universe, relational time, many-worlds interpretation, configuration space geometry, Fubini–Study metric, scalar curvature, scalar coupling, atomic clocks, quantum foundations

I. INTRODUCTION

Quantum mechanics admits formulations in which time is not fundamental, notably constraint-based and timeless approaches. Yet physical experience exhibits strong temporal ordering and persistent correlations. Dynamic Location Theory proposes that such correlations emerge from weak couplings between nearby configurations in quantum state space rather than from linear time evolution alone.

Previous formulations introduced a scalar coupling γ phenomenologically. Here, we derive γ from first principles by treating quantum state space geometrically. The central claim is that the curvature of configuration space induces suppressed but nonzero inter-configuration correlations, manifesting as temporal continuity. This approach provides a natural, dimensionally consistent explanation for emergent temporal behavior.

II. CONFIGURATION SPACE GEOMETRY

A. Projective Hilbert Space

Let $\mathcal{H}$ be a complex Hilbert space with normalized pure states $|\psi\rangle$. Physical states correspond to rays, forming the projective Hilbert space $\mathcal{P}(\mathcal{H})$. This identification removes the physically irrelevant global phase.

B. Fubini–Study Metric

For a smoothly parameterized state $|\psi(\lambda^\mu)\rangle$, the infinitesimal distance must be invariant under global phase changes. The squared line element is

$$ds^2 = \langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2, \quad (1)$$

which subtracts the contribution from pure phase variation. Equivalently, the metric tensor is

$$g_{\mu\nu} = \Re [\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]. \quad (2)$$

This metric defines the infinitesimal distinguishability of quantum states in parameter space, forming the foundation for curvature calculations.

III. CURVATURE OF QUANTUM STATE SPACE

A. Geometric Objects

From $g_{\mu\nu}$, we construct the Levi–Civita connection, Riemann tensor, Ricci tensor, and scalar curvature R using standard differential geometry.

B. Single Qubit

Consider

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (3)$$

Derivatives:

$$\partial_\theta |\psi\rangle = -\frac{1}{2} \sin \frac{\theta}{2} |0\rangle + \frac{1}{2} e^{i\phi} \cos \frac{\theta}{2} |1\rangle, \quad (4)$$

$$\partial_\phi |\psi\rangle = ie^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (5)$$

Inner products yield

$$g_{\theta\theta} = \frac{1}{4}, \quad g_{\phi\phi} = \frac{1}{4} \sin^2 \theta. \quad (6)$$

Thus

$$ds^2 = \frac{1}{4} (d\theta^2 + \sin^2 \theta d\phi^2), \quad (7)$$

which is the round metric on S^2 of radius $1/2$. The scalar curvature is

$$R = 8. \quad (8)$$

(Note: Alternative normalizations scale R by a constant factor; here we adopt the quantum-information convention yielding $R = 8$.)

C. Two-Qubit Product States

For a product state $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, the metric decomposes additively:

$$ds_{\text{prod}}^2 = ds_1^2 + ds_2^2, \quad R_{\text{prod}}^{(2)} = 8 + 8 = 16. \quad (9)$$

By induction, for N -qubit product states:

$$R_{\text{prod}}^{(N)} = 8N. \quad (10)$$

D. Entangled Two-Qubit States

For a Schmidt-rank-2 state

$$|\Psi\rangle = \cos \alpha |00\rangle + e^{i\beta} \sin \alpha |11\rangle, \quad (11)$$

the metric is

$$ds^2 = d\alpha^2 + \sin^2 \alpha d\beta^2, \quad (12)$$

with scalar curvature

$$R(\alpha) = 8 - 24 \sin^2(2\alpha). \quad (13)$$

Maximum entanglement ($\alpha = \pi/4$): $R_{\text{Bell}} = -16$. Product limit: $R_{\text{sep}} = 8$. Entanglement can flip the sign of curvature.

E. N -Qubit Entangled Manifolds

For GHZ-type states

$$|\Psi_N\rangle = \cos \alpha |0\rangle^{\otimes N} + \sin \alpha |1\rangle^{\otimes N}, \quad (14)$$

the effective 2D parameter space curvature scales inversely:

$$R_{\text{GHZ}}(N) \sim -\frac{c}{N}, \quad c = \mathcal{O}(1). \quad (15)$$

Dimensional collapse of accessible configuration space implies suppression with system size ($\beta = 1$).

IV. EMERGENCE OF THE SCALAR COUPLING γ

A. Effective Interaction

We model γ as an effective coupling:

$$S_\gamma = \gamma \int d^4x \phi(x) \rho(x), \quad (16)$$

where $\phi(x)$ encodes configuration space proximity, and $\rho(x)$ is the local energy density.

B. Dimensional Consistency

Given $[\rho] = \text{energy} \cdot \text{length}^{-3}$, $[d^4x] = \text{length}^4$, $\gamma\phi$ must have dimensions $(\text{energy} \cdot \text{length})^{-1}$. Fubini–Study curvature R is dimensionless, so a fundamental scale is needed. We introduce the Planck length ℓ_P and mass M_P .

C. Definition of γ

We define

$$\gamma = \alpha \frac{|\langle R \rangle|}{M_P^2 \ell_P^2}, \quad (17)$$

with $\alpha \sim 1$ and $\langle R \rangle$ the averaged scalar curvature. This ensures correct units and Planck suppression. In path-integral formulations over curved state space, γ effectively arises as $\exp(-R/R_0)$, with $R_0 = 8$ from the qubit baseline.

D. Many-Body Scaling

For N particles:

$$\gamma(N) = \frac{\alpha R_1}{N^\beta M_P^2 \ell_P^2}, \quad \beta = 1, \quad (18)$$

showing suppression for large systems and enhancement for small, coherent quantum systems.

V. APPLICATION TO ATOMIC CLOCKS

A. Simplified Model

Consider an effective two-level clock Hamiltonian

$$\hat{H} = \omega_0 \sigma_z + \Omega(t)(\sigma_+ + \sigma_-). \quad (19)$$

The accessible state manifold is a deformed Bloch sphere; curvature depends on driving and decoherence.

B. Predicted γ Values

Single-ion clocks ($N = 1$): $\gamma \sim 10^{-16}$ – 10^{-17} . Optical lattice clocks ($N \sim 10^4$): $\gamma \lesssim 10^{-20}$. These values are below current experimental bounds, consistent with null results from clock comparisons.

VI. OBSERVABLE PREDICTIONS

1. ****Entanglement Scaling****: $\Delta f/f \propto N^{-\beta}$, with $\beta = 1$. Deviations falsify the model.
2. ****Geometric Phase Anomaly****: Closed loops in parameter space yield an additional phase proportional to γ and integrated curvature, distinct from standard Berry phase effects.
3. ****Correlation Signatures****: Weak frequency correlations emerge when spatially separated clocks trace geometrically proximate trajectories in state space.

VII. DISCUSSION

The scalar coupling γ arises naturally as a geometric invariant, suppressed by the Planck scale and modulated by many-body structure. Temporal continuity emerges from configuration space proximity rather than fundamental time flow. This unifies quantum geometry, information-theoretic distance, and observable physics. Potential connections to quantum gravity, entanglement entropy, and holographic curvature merit future exploration.

VIII. CONCLUSION

We developed a fully geometric Dynamic Location Theory, deriving γ from first principles. The theory is dimensionally consistent, bounded by experimental constraints, and falsifiable. Future work includes connections to quantum gravity, information geometry, and path integral formulations over curved quantum state spaces.

ACKNOWLEDGMENT

The author thanks the broader physics community for continued dialogue on the foundations of time and quantum mechanics.

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